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Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2024

Time: 03 Hours

Code: 101406 ***Subject: Materials, Testing & Evaluation*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only):- **[14]**

(a) Slump test for concrete is carried out to determine

(i) Strength (iii) Workability
(ii) Durability (iv) Water content

(b) Which of the following is an example of a thermoplastic?

(i) Melamine (iii) Epoxide
(ii) Urethane (iv) Acetal

(c) Consider the following statements regarding the advantages of plywood:
A. It has good strength both along as well as across the grains.
B. It will not shrink or swell across the grains.
C. It can be curved into desired shapes.
Which of the above statements is/are correct?

(i) A and C (iii) B and C
(ii) A and B (iv) B only

(d) A cylindrical bar of 25 mm diameter and 1.2 m length is subjected to a longitudinal strain during a tensile test is four times the lateral strain, what is the value of shear modulus? (Take the modulus of elasticity as 1.2×10^5 N/mm^2)

(i) 5.8 kN/mm^2 (iii) 58 kN/mm^2
(ii) 4.8 kN/mm^2 (iv) 48 kN/mm^2

(e) Which one of the following statements is not correct for Aluminium?

(i) It is less ductile than copper (iii) It can be soldered
(ii) It is harder than tin (iv) It can be welded

(f) The drawback of electrical seasoning of timber is

(i) Checks (iii) Cracks
(ii) Splitting (iv) Reduced strength

(g) Gap grading is one

(i) in which the particles fall within a narrow limit of size fractions
(ii) In which one or more intermediate fractions are absent
(iii) Which combines different fractions of fine and coarse aggregates
(iv) In which all the particles are of uniform size

(h) Which of the following class of engineering ceramics generally included lubricant materials?

(i) Metalloids (iii) Sulphides
(ii) Intermetallics (iv) Carbides

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(i) Which of the following is not a method of fabrication of rubber?

(i) Vacuum forming
(ii) Buffing
(iii) Calendering
(iv) Compression molding

(j) Polymer materials become brittle below

(i) Critical temperature
(ii) Glass transition temperature
(iii) Melting point temperature
(iv) Room temperature

Q.2 Solve both questions : **[14]**

(a) What is meant by the term refractories? Mention the desired properties of refractories. Also state the properties and uses of acidic, basic and neutral refractories.

(b) Explain the tests to be conducted to check the suitability for the construction work.

Q.3 Solve both questions : **[14]**

(a) Enumerate the laboratory tests for cement and describe any two of them.

(b) What are the objectives of preservation of timber? Describe the preservatives which are commonly used in the process of preservation of timber.

Q.4 Solve both questions : **[14]**

(a) Describe manufacturing process of pig-iron by blast furnace.

(b) Enumerate various markets form of steel with neat sketches.

Q.5 Solve both questions : **[14]**

(a) State the properties of aluminium and also discuss the importance of aluminium as structural material.

(b) How is coloured glass made? State the colouring substances which are used for getting different shades of colour of glass.

Q.6 Solve both questions : **[14]**

(a) Explain briefly various testing procedure for reinforcing steel bars.

(b) What is fracture mechanics. How and why fracture toughness tests are performed in laboratory?

Q.7 Solve both questions : **[14]**

(a) The results of sieve analysis of 600 gm oven dried soil sample are shown below:

Sieve No.	Diameter (mm)	Mass of soil retained on sieve (g)
4	4.75	28
10	2	42
20	0.85	48
40	0.425	121
60	0.25	211
100	0.15	86
200	0.075	40
Pan		24

Determine the percent finer than each sieve and plot a grain size distribution curve.

(b) Explain factors affecting ductile to brittle transition temperature (DBTT).

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Q.8 Solve both questions : **[14]**

(a) Define structural integrity and discuss briefly various procedures for structural integrity assessment with examples.

(b) What is the difference between True stress-strain and Engineering stress-strain curve? Show with neat sketches. Which one is implemented in design codes and why?

Q.9 Write short notes on any four of the following:- **[14]**

(a) Flash point and fire point of bitumen

(b) Annealing of glass

(c) Geo-textiles

(d) Acoustic material

(e) Plastic deformation of metals

(f) Graphene

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